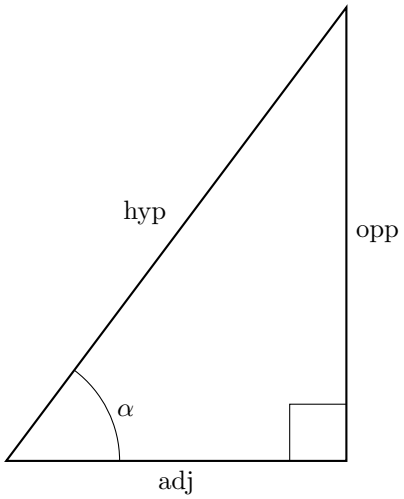
MATH 195: 10/22
RIGHT TRIANGLE TRIGONOMETRY

Trigonometry in quadrant I, with angles between 0 and $\pi/2$, can be interpreted as being about right triangles.

- $\cos(\alpha) = \dfrac{\text{adj}}{\text{hyp}}$
- $\sin(\alpha) = \dfrac{\text{opp}}{\text{hyp}}$
- $\tan(\alpha) = \dfrac{\text{opp}}{\text{adj}}$



The other three trig functions are reciprocals of these, or you can remember them in terms of ratios of sides.

- $\sec(\alpha) = \dfrac{1}{\cos(\alpha)} = \dfrac{\text{hyp}}{\text{adj}}$
- $\csc(\alpha) = \dfrac{1}{\sin(\alpha)} = \dfrac{\text{hyp}}{\text{opp}}$
- $\cot(\alpha) = \dfrac{1}{\tan(\alpha)} = \dfrac{\text{adj}}{\text{opp}}$

You can think of each of these as an equation relating two sides and one angle. If you know two of the three unknowns you can solve for the third.

Co-function identities.
A right triangle has two angles you can use as inputs for trig functions. Thinking about this gives you relationships between trig functions, giving a sense in which they are opposites of each other.

$$\sin(\pi/2 - \alpha) = \cos\alpha \qquad \cos(\pi/2 - \alpha) = \sin(\alpha)$$
$$\sec(\pi/2 - \alpha) = \csc\alpha \qquad \csc(\pi/2 - \alpha) = \sec(\alpha)$$
$$\tan(\pi/2 - \alpha) = \cot\alpha \qquad \cot(\pi/2 - \alpha) = \tan(\alpha)$$

Inverse trig functions.
Restricted to quadrant I, the interval $(0, \pi/2)$, every trig function is one-to-one. Thus they have inverses, written either trig^{-1} or arctrig. That is, if α is in quadrant I then

- $\cos(\alpha) = x$ is equivalent to $\arccos(x) = \alpha$
- $\sin(\alpha) = y$ is equivalent to $\arcsin(y) = \alpha$
- $\tan(\alpha) = m$ is equivalent to $\arctan(m) = \alpha$

You can think of these as saying, if you know the ratio of side lengths you can determine the angle of the triangle.

You can talk about inverse trig functions over a larger domain than quadrant I, but things get more complicated. The challenge is that the trig functions are not one-to-one on larger domains. We will come back to this question after we've looked at graphs of trig functions.

Practice problems

For all questions, either give an exact answer or round to two digits after the decimal point. If you use a calculator, make sure you have to set to use radians or degrees as appropriate for each problem. For all problems, if you are not sure how to proceed start by drawing a picture.

(1) A right triangle has an angle of $30°$ and a hypotenuse of 10. Determine the lengths of the other two sides.
(2) A right triangle has an angle of $\pi/4$ with the leg adjacent to that angle being 4. Determine the lengths of the other two sides.
(3) A right triangle has legs of length 2 and 6. Determine the length of the hypotenuse and the angles of the triangle.
(4) A right triangle has a hypotenuse of length 13 and a leg of length 5. Determine the length of the other leg and the angles of the triangle.
(5) Measuring heights of objects can be difficult. A device known as a clinometer, for measuring angles, combined with trigonometry can help. Namely, you measure a distance d from the base of the object, and then use a clinometer to measure the angle α from that distance to the top of the object. Come up with a formula using a trig function which gives you the height h based on d and α.
(6) Apply your formula for the following case: you measure that 20 meters away from a flagpole the angle to the top of the pole is $40°$. Determine the height of the flagpole.
(7) Apply your formula for the following case: you measure that 12 feet away from a tree the angle to its top is $65°$. Determine the height of the tree.
(8) A funicular railway provides transportation up a steep slope. You measure the incline of the slope is $35°$ above the horizon. If the distance of the railway along the slope is 241 meters determine the elevation gain from bottom to top.
(9) Explain why $\arccos x$ and $\arcsin x$ are undefined when $x > 1$.
(10) Your friend says that $\arctan x$ is undefined when $x > 1$. If they are correct, explain why. If not, determine which positive values of x have $\arctan x$ defined, and explain.
(11) Many calculators don't have buttons for secant, cosecant, and cotangent. Explain how to calculate them using only the buttons for cosine, sine, and tangent.
(12) Most calculators don't have buttons for arcsec, arccsc, and arccot. Explain how to calculate them using only the buttons for arccos, arcsin, and arctan.
(13) Draw your own right triangle with partial information about its sides and angles. Use trig to solve for the remaining sides and angles.
(14) What is the minimum number of pieces of info needed to solve for the remaining info about a right triangle?
(15) Write your own word problem about using right triangle trig to calculate something. Solve your problem.